DECENCY DESCENDS ALONG A QUASI-COMPACT SURJECTIVE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterisation of surjectivity of a morphism of algebraic stacks through the surjectivity on points, and the criterion for decency through the quasi-compactness of a fibre product along a presentation. Invoked inside the counted proof (source lines 10473-10480) as pointers to results of the chapter on properties of stacks and of this chapter. Reduces the assertion to the quasi-compactness of a fibre product over a field valued point.
- The quasi-compactness of the morphism from a decent stack to a base along a field valued point, and the stability of quasi-compactness of a fibre product under such a point. Used inside the counted proof (source lines 10481-10489). Provides the two quasi-compact spaces whose fibre product computes the required one.

Source reference. Original lemma lemma-check-quasi-compact-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2243–2243. Navigation pointer only: the counted text refers to the Stacks item lemma-check-quasi-compact-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-characterize-surjective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4141–4141. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-characterize-surjective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If f is quasi-compact and surjective and $\mathcal{X}$ is decent, then $\mathcal{Y}$ is decent.*

Proof. Let $x : \mathrm{Spec}(k) \to \mathcal{X}$ be a morphism where k is a field and denote $y = f \circ x$. Since f is surjective, every point of $|\mathcal{Y}|$ arises in this manner, see Properties of Stacks, Lemma stacks-properties-lemma-characterize-surjective. Consider an affine scheme T and morphism $T \to \mathcal{Y}$. It suffices to show that $T \times_{\mathcal{Y},y} \mathrm{Spec}(k)$ is quasi-compact, see Lemma lemma-check-quasi-compact-covering. We have

$$(T \times_{\mathcal{Y}} \mathcal{X}) \times_{\mathcal{X},x} \mathrm{Spec}(k) = T \times_{\mathcal{Y},y} \mathrm{Spec}(k)$$

The morphism $T \times_{\mathcal{Y}} \mathcal{X} \to T$ is quasi-compact hence $T \times_{\mathcal{Y}} \mathcal{X}$ is quasi-compact. Since x is a quasi-compact morphism as $\mathcal{X}$ is decent we see that the displayed fibre product is quasi-compact. □